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PinCFlow.jl: An idealized-atmospheric-flow solver coupled to the 3D transient gravity-wave model MS-GWaM

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Summary

PinCFlow.jl (*P*seudo-*in*Compressible *Flow* solver) is an atmospheric-flow solver that was designed for conducting idealized simulations. It integrates the Boussinesq, pseudo-incompressible and compressible equations in a conservative flux form (Klein, 2009; Rieper et al., 2013), using a semi-implicit method that combines explicit and implicit time-stepping schemes (Benacchio & Klein, 2019; Chew et al., 2022; Schmid et al., 2021). Spatially, the equations are discretized with a finite-volume method, such that all quantities are represented by spatial averages over grid cells and fluxes are computed on the respective cell interfaces. The grid is staggered (Arakawa & Lamb, 1977) so that the velocity components are defined at the same points as the corresponding fluxes of scalar quantities. PinCFlow.jl operates in a vertically stretched terrain-following coordinate system based on Gal-Chen & Somerville (1975a), Gal-Chen & Somerville (1975b) and Clark (1977).

The Lagrangian gravity-wave parameterization MS-GWaM (*M*ulti-*S*cale *G*ravity-*W*ave *M*odel) is interactively coupled to the dynamical core of PinCFlow.jl, so that unresolved gravity waves may be parameterized in a manner that accounts for transient wave-mean-flow interaction and horizontal wave propagation. The resolved fields are updated with tendencies computed by MS-GWaM at the beginning of every time step. A description of the theory behind MS-GWaM can be found in Achatz et al. (2017) and Achatz et al. (2023). For a numerical perspective and more information on the development, see Muraschko et al. (2015), Bölöni et al. (2016), Wilhelm et al. (2018), Wei et al. (2019) and Jochum et al. (2025).

Statement of need

Atmospheric gravity waves are generated by various sources, including convection, flow over mountains and jet-frontal imbalances. As they propagate upward, they transport and deposit momentum and energy, thus impacting the large-scale flow. A significant part of the gravity-wave spectrum cannot be resolved by weather and climate models at their respective standard resolutions and must therefore be parameterized (Achatz et al., 2024; Alexander et al., 2010; Fritts & Alexander, 2003).

Most gravity-wave parameterizations rely on single-column and steady-state approximations, since these yield a comparatively simple and efficient method of calculating the mean-flow impact. However, it has become increasingly apparent that transient wave-mean-flow interaction and horizontal gravity-wave propagation need to be accounted for in an accurate description of gravity-wave-mean-flow interaction (Bölöni et al., 2016; Ehard et al., 2017; Muraschko et al.,

2015; Sato et al., 2009; Senf & Achatz, 2011; Wei et al., 2019). The Multi-Scale Gravity-Wave Model (MS-GWaM) has been created for this reason. Essential to its development, as to the improvement of the fundamental understanding of atmospheric dynamics, is the study of gravity waves in idealized simulations. PinCFlow.jl provides a modeling infrastructure tailored to this exact need. Moreover, it is interactively coupled to MS-GWaM, thus allowing for direct comparison between high-resolution wave-resolving simulations and coarse-resolution wave-parameterizing simulations.

Features

Figure 1 shows an overview of PinCFlow.jl's features. As has been mentioned above, the model can be used to integrate the Boussinesq, pseudo-incompressible or compressible equations. For each of these, it provides a number of background atmospheres defined by model parameters. The initial deviations of the prognostic variables from the background state are set with user-defined functions. In addition to the momentum, continuity and potential-temperature equations, the flow solver integrates the prognostic equation for an arbitrary passive tracer with direct gravity-wave impact parameterized using MS-GWaM according to Knop et al. (2025).

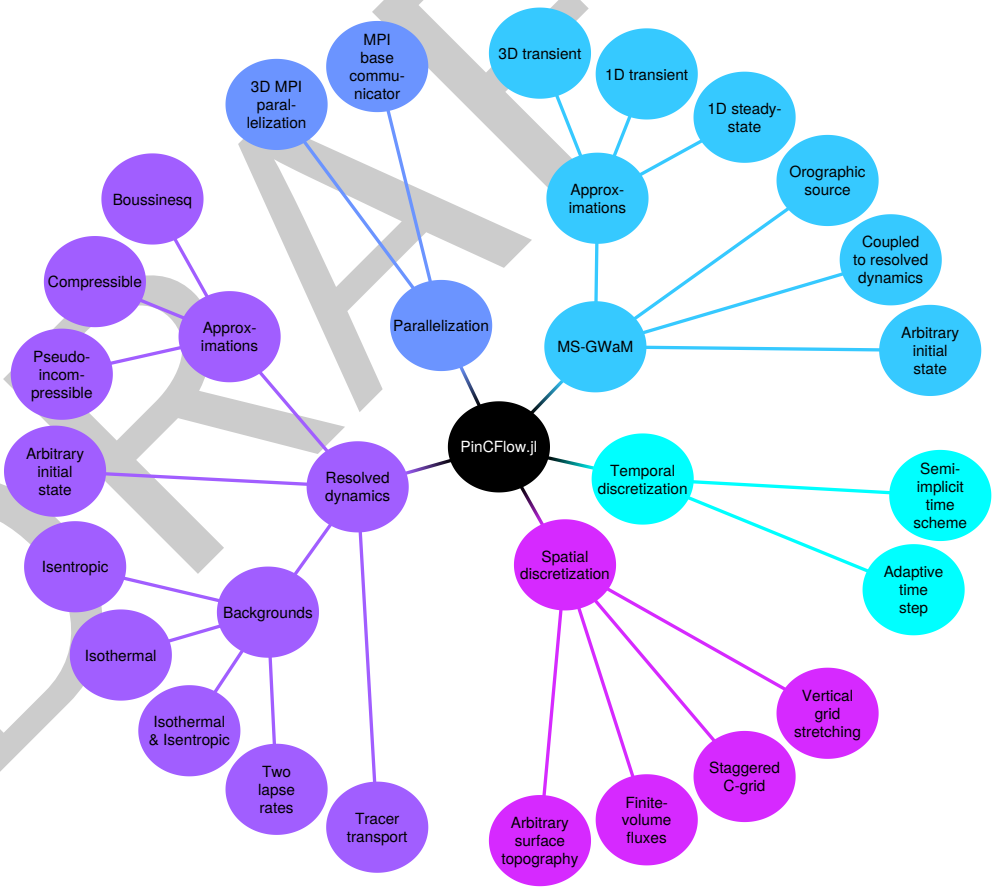

Figure 1: Overview of PinCFlow.jl's features.

PinCFlow.jl's grid is determined from a user-defined surface topography, using a height-based terrain-following coordinate system. Fluxes are computed with a finite-volume (MUSCL) scheme (e.g. Leer, 2003), using the monotonized-centered variant limiter (e.g. Kemm, 2011). In addition to the advection terms, the model includes molecular friction, heat conduction and

a turbulent viscosity.

MS-GWaM can be run in a 1D steady-state mode (which represents classic gravity-wave parameterizations), a 1D transient mode (so that the impact of transient wave-mean-flow interaction may be investigated separately, i.e. without horizontal propagation) and a 3D transient mode (the full implementation of the theory). It is interactively coupled to the resolved dynamics, so that the evolution of the gravity-wave field depends on the current state of the large-scale flow and in turn changes it via tendencies that are added to the prognostic equations. This coupling includes the impact of gravity-wave fluxes on the large-scale tracer. The parameterization has an inbuilt orographic source which launches mountain waves generated by the resolved flow passing over unresolved terrain. The latter is represented by an arbitrary spectrum computed by a user-defined function. Moreover, MS-GWaM may be initialized with any wave field, also via a user-defined function.

By default, the flow solver uses an adaptive time step determined from several stability criteria, including CFL conditions with respect to the resolved flow and the maximum group velocities of the unresolved gravity-wave field computed by MS-GWaM. The CFL numbers are configurable, however, one may also run the model with a fixed time step.

MPI parallelization is supported in all grid dimensions. The number of MPI tasks is set per dimension, using corresponding model parameters. Furthermore, one may pass a base communicator to the model, facilitating the setup of ensemble simulations in a single Julia script.

Related research and software

PinCFlow.jl originated from a Fortran code (Rieper et al., 2013) that was used and extended for gravity-wave research (Bölöni et al., 2016; Jochum et al., 2025; Schmid et al., 2021; Wei et al., 2019; Wilhelm et al., 2018). Its semi-implicit time scheme (Schmid et al., 2021) is based on the temporal discretization in the models introduced by Benacchio & Klein (2019) and Chew et al. (2022). In contrast to these models, PinCFlow.jl does not provide seamless access to soundproof and hydrostatic dynamics (instead, the prognostic equations are set in the initialization and not changed during the simulation) or any data assimilation capabilities. What distinguishes it from other software is its implementation of MS-GWaM. The latter was originally developed as an isolated code (Bölöni et al., 2016; Muraschko et al., 2015) and later integrated into the predecessor of PinCFlow.jl (by Jochum et al., 2025; Wei et al., 2019; Wilhelm et al., 2018) and the NWP-and-climate model ICON (by Bölöni et al., 2021; Y.-H. Kim et al., 2021; Y.-H. Kim et al., 2024; Voelker et al., 2024). PinCFlow.jl's version of MS-GWaM differs from the ICON version in several components, most notably its coordinate system, discretization and sources, however, the underlying theory (Achatz et al., 2017, 2023) is the same. The code presented here provides the first open access to a full implementation of MS-GWaM coupled to a flow solver within an idealized and highly configurable framework.

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